Bayesian Updating of Bridge Structural Reliability Using Monitoring Data, Stop Criteria, and Proof Load Testing: A Case Study of the Zijlweg Viaduct

Extended Abstract and Introduction

Neguse Solomon Mekonnen · MSc dissertation, International Master in Risk Assessment and Management of Civil Infrastructures (NoRISK), UPC Barcelona / University of Padua, 2026

Supervisors: Dr. Eva Lantsoght and Dr. Rolando Chacón. ***A journal manuscript based on this work is in preparation.***

Extended Abstract

Aging reinforced concrete bridges face a recurring assessment problem: deterioration such as alkali-silica reaction (ASR) makes analytical capacity models unreliable, so structures that may be adequate in reality can appear insufficient on paper. Proof load testing offers direct experimental evidence of adequacy, but its value is underused when the outcome is read only as pass or fail. This dissertation shows how the monitoring data recorded during a proof load test can be turned into quantitative reliability evidence.

I applied the Bayesian reliability updating framework of De Vries et al. to field data from the 2015 proof load test of the Zijlweg viaduct, a 1965 ASR-affected reinforced concrete solid slab bridge in Noord-Brabant, the Netherlands. The bridge was loaded at two positions, one critical for bending and one for shear, over five load levels up to 1,342 kN. Concrete strain (bending) and maximum weighted nominal crack width (shear) were used to update the resistance distribution at each load step, yielding the annual reliability index before, during and after the test. In parallel, mechanics-based stop criteria derived from the Critical Shear Crack Theory (CSCT) and the Critical Shear Displacement Theory (CSDT), arranged as a traffic-light system, were evaluated on the same data. The updating relies on Metropolis-Hastings MCMC (1×10^8 samples per load step), a Latin Hypercube cross-section analysis with 100 realisations, and a MATLAB preprocessing workflow.

The corrected cross-section analysis, including all four layers of bottom reinforcement from the drawings, gives a mean yield moment of 3,248.87 kNm with a coefficient of variation of 0.137. After the full test the posterior reliability index is $\beta = 5.33$ for bending and $\beta = 4.89$ for shear, both above the annual target for new structures ($\beta = 4.7$). The annual target for existing structures at the disapproval level ($\beta = 4.0$) is reached after the third load step for bending ($\beta = 4.40$) and the fourth for shear ($\beta = 4.41$). At the third shear load step, the monitored crack-width information gives $\beta = 3.81$ against 1.96 for the survival-only lower-bound approach, an advantage of 1.85 reliability-index units. The CSCT/CSDT indicators stayed in the green range throughout the shear test, consistent with the positive reliability results at every step.

The results indicate that an instrumented proof load test, interpreted through a Bayesian framework, can serve as a Level IV assessment for an ASR-affected bridge that looks insufficient under lower assessment levels. Stop criteria and Bayesian updating play complementary roles: the first gives an operational safety signal during the test, the second a quantitative reliability estimate after it.

1. Introduction

1.1 Context and motivation

Much of Europe's bridge stock was built decades ago and is now exposed to deterioration, growing traffic and higher expectations of safety. Code-based assessment applies design-oriented rules to existing structures, ignores system effects and material overstrength, and cannot reflect the condition revealed by inspection. The result can be unnecessary load restrictions, strengthening or replacement. Probabilistic assessment improves on this by expressing safety through reliability indices, but it often relies on generic models that may not describe the individual structure.

1.2 Proof load testing and monitoring data

A proof load test loads a bridge to a level representative of factored design loads to demonstrate that it can carry them without significant distress. It is most valuable when documentation is incomplete, when deterioration undermines analytical models, or when analysis indicates insufficient capacity that the structure may not actually have. All three conditions apply to the Zijlweg viaduct, which shows ASR and was rated insufficient for shear analytically. Modern tests are heavily instrumented, and each load level that the structure sustains is evidence about its true capacity. Bayesian methods allow that evidence to be combined formally with prior knowledge, so that monitoring data reduce uncertainty about resistance.

1.3 Research gap

Bayesian reliability updating for proof load testing has been demonstrated by De Vries et al. on laboratory slab strips with an analytical flexural model. Mechanics-based stop criteria for shear-critical slabs have been validated separately. As far as the dissertation's literature review shows, the framework had not been applied to field monitoring data from a real ASR-affected viaduct, and its reliability estimates had not been compared load step by load step with CSCT/CSDT stop criteria.

1.4 Research question and objectives

The central question is how monitoring data from an instrumented proof load test on an ASR-affected reinforced concrete bridge can be used to update structural reliability quantitatively, and how those Bayesian estimates compare with stop criteria evaluated at each load step. The specific objectives are to extract and verify the indicator values from the Zijlweg data and apply them in the updating framework; to compare the updated reliability with the lower-bound approach at each load step; and to assess whether the stop-criteria signals agree with the reliability-based assessment.

1.5 Approach, scope and contribution

The dissertation adopts the existing framework of De Vries et al. and its C++ implementation, and its contribution is the application to field data. This included data preprocessing in MATLAB, derivation of the statistical inputs through a probabilistic cross-section analysis, adaptation of the existing C++ codes to

the test's measurement parameters and loading conditions, and the integration of stop-criteria evaluation alongside the updating. Bending and shear were analysed separately, with a focus on ultimate limit states and a single case study.

The main contributions are:

1. An application of the framework to real field proof load test data from an ASR-affected bridge;
2. A quantification of the advantage of monitoring data over the survival-only lower bound;
3. A direct comparison between the Bayesian estimates and CSCT/CSDT stop criteria in an actual test;
4. A demonstration that omitting reinforcement layers in the sectional model can reduce the calculated yield moment by up to 44% and distort the reliability results; and
5. A documented case study, with the preprocessing workflow and verified inputs, that can serve as a reference for later applications. The limits of the study are its single case, its dependence on the 2015 test's instrumentation and loading protocol, its assumption of stationary material properties during the test, and its use of existing algorithms with modification rather than new methods.